

LM14 Credit Risk

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1. Introduction

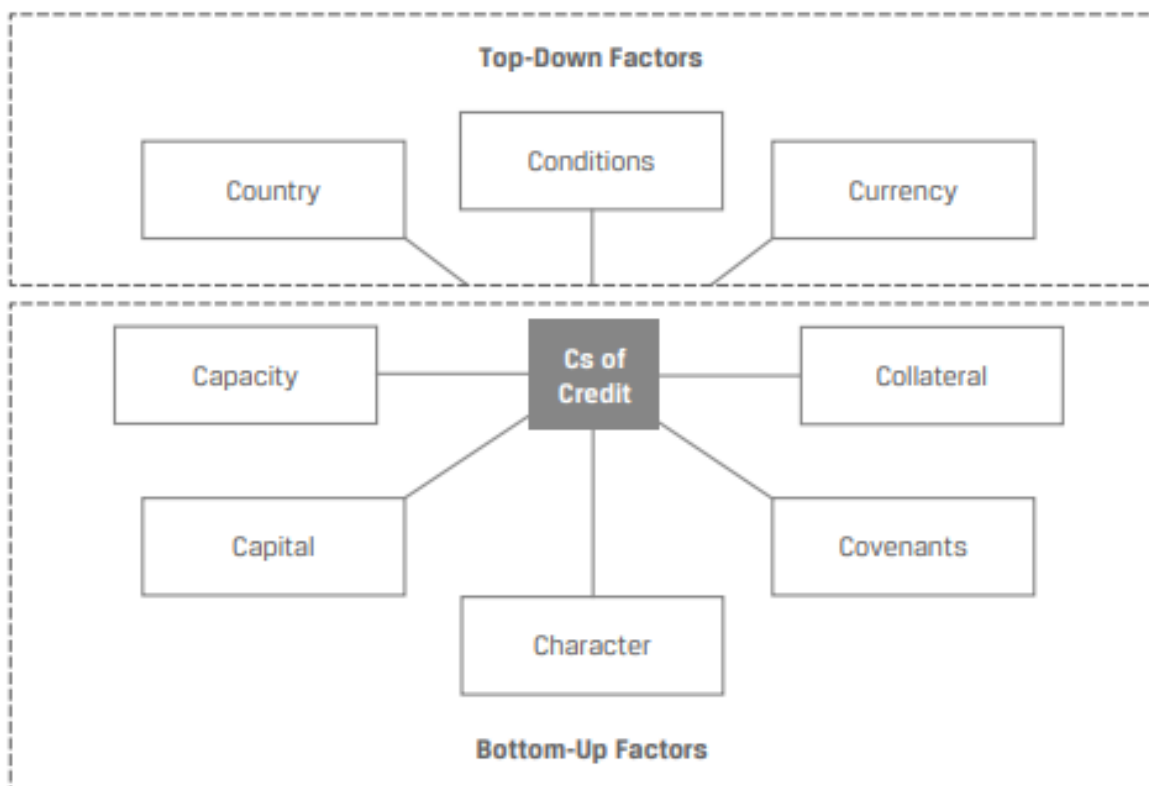
This learning module covers:

- Credit risk and its components – probability of default and loss given default
- Role of credit ratings in debt markets
- Factors that influence the level and volatility of yield spreads

2. Sources of Credit Risk

Credit risk is the risk that the borrower will fail to make principal and/or interest payments on time.

Traditionally, many analysts assessed creditworthiness using what are known as the "Cs of credit analysis," as illustrated in Exhibit 1:



The five bottom-up factors that are applicable to an individual borrower are:

- 'Capacity' refers to the ability of the borrower to make its debt payments on time.
- 'Capital' refers to other company resources available that reduce reliance on debt.
- 'Character' refers to the quality of the management and the willingness to pay debt.
- 'Covenants' refers to the terms and conditions of the debt agreement that the issuer

must comply with.

- 'Collateral' refers to the quality and value of the assets pledged against the debt.

The remaining three factors are general top-down factors that apply to all borrowers:

- 'Conditions' refers to general economic, competitive, and business environment faced by all borrowers that may impact their ability to service or refinance debt.
- 'Country' refers to the geopolitical environment as well as the legal and political system faced by all issuers in a jurisdiction.
- 'Currency' refers to cash flows affected by exchange rate changes, such as import/export or borrowings in foreign currency.

Sources of Credit Risk

Exhibit 3 from the curriculum shows the main borrower types, their primary sources of repayment and credit risk.

Borrower Type	Primary Source of Cash Flow Generation	Secondary Source of Cash Flow Generation	Sources of Credit Risk
Corporate	▪ Business operations ▪ Investment and financing activities	▪ Asset sales ▪ Divestitures ▪ Additional debt issuance ▪ Equity issuance	▪ Economic contraction ▪ Strategic shifts in the business and market environment ▪ Increased competition ▪ Reduced pricing power ▪ Shrinking operating margins, increased losses ▪ Excessive debt service needs
Sovereign or public entity	▪ Corporate and personal income taxes, sales tax and VAT revenue, capital gains and wealth-based taxes ▪ Tariffs, fees, and other government revenue	▪ Newly issued debt ▪ Sale of public assets, privatization	▪ Economic contraction ▪ Political uncertainty ▪ Excessive debt service needs ▪ Expansionary economic policies ▪ Budget deficits ▪ Tax cuts ▪ Limited ability to collect taxes

As shown in the exhibit, many factors can affect the primary and secondary sources of repayment for different fixed-income issuers.

Measuring Credit Risk

Credit risk has two components:

- Probability of default (POD): The likelihood that an issuer fails to make full and timely payments of principal and interest.
- Loss given default (LGD): How severe is the loss incurred by the investor? What

portion of the bond's value (including interest) is not paid by the issuer? A default leads to various severity of losses; for instance, the loss may be total, or the bondholders may recover some value.

We combine both the components into a single term called the 'expected loss'.

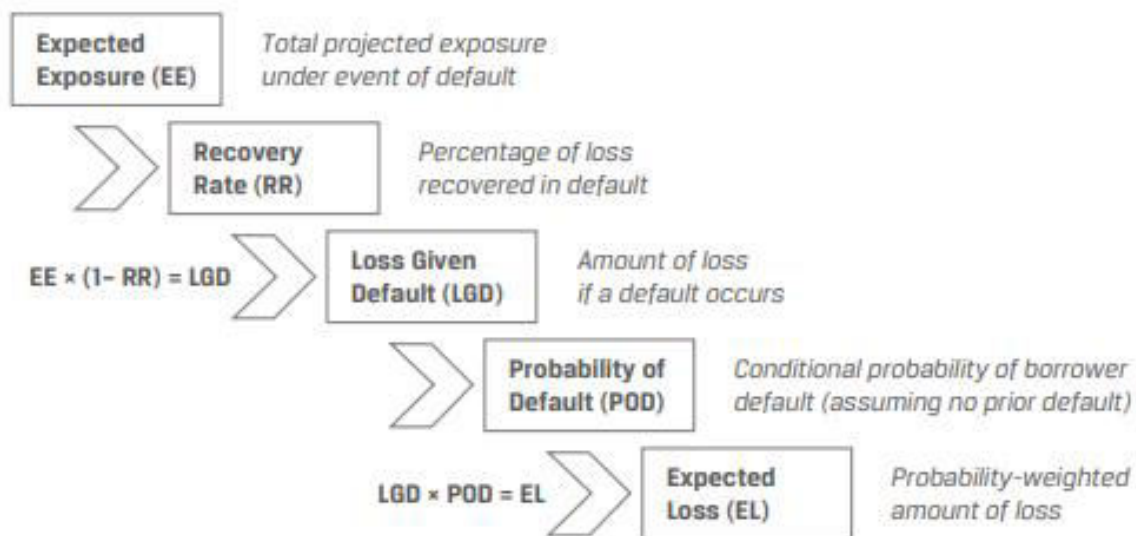
$$\text{Expected Loss} = \text{POD} \times \text{LGD}$$

The loss given default depends on:

- Exposure at default (EAD): The size of the investor's claim at the time of default.
- Recovery rate (RR): Represents the percentage of an outstanding debt claim recovered when an issuer defaults. A related term is loss severity (1-RR) which represents the unrecovered portion of the claim.

$$\text{LGD} = \text{EE} \times (1 - \text{RR})$$

Exhibit 4 from the curriculum illustrates these concepts:



One way to interpret a fixed-income security's expected loss for a given period is to compare it to the compensation an investor expects for taking on a borrower's credit risk over that period, which is the credit spread. We can say that an investor is fairly compensated if the expected loss is equal to the credit spread for a given period.

$$\text{Credit Spread} \approx \text{POD} \times \text{LGD}$$

Example:

(This is based on Question Set – Q#4 from the curriculum.)

A bond investor analyzing a company's unsecured debt estimates a POD of 2% and an LGD of 80%. The observed actual credit spread for this bond is 200 bps per year. Is the investor fairly compensated/ less than fairly compensated/more than fairly compensated for

assuming the company's credit risk?

Solution:

Since $\text{POD} \times \text{LGD} = 1.60\%$ and the credit spread is 2.00% ,

Credit Spread $>$ $\text{POD} \times \text{LGD}$ and the investor would expect to be more than fairly compensated for assuming the company's credit risk.

3. Credit Rating Agencies and Credit Ratings

Moody's, S&P, and Fitch are the three main rating agencies. At least two of these agencies assign a rating to the majority of the bonds.

Credit Ratings

Be familiar with what the ratings are (Aaa represents the highest quality among investment grade securities, etc.). The rating system is almost similar across agencies starting from A for investment grade to C/D for junk.

Long-Term Ratings Matrix: Investment Grade vs. Non-Investment Grade

		Moody's	S&P	Fitch
Investment Grade	High-Quality Grade	Aaa	AAA	AAA
		Aa1	AA+	AA+
		Aa2	AA	AA
		Aa3	AA-	AA-
	Upper-Medium Grade	A1	A+	A+
		A2	A	A
		A3	A-	A-
	Low-Medium Grade	Baa1	BBB+	BBB+
		Baa2	BBB	BBB
		Baa3	BBB-	BBB-

		Moody's	S&P	Fitch
Non-Investment Grade "Junk" or "High Yield"	Low Grade or Speculative Grade	Ba1	BB+	BB+
		Ba2	BB	BB
		Ba2	BB-	BB-
		B1	B+	B+
		B2	B	B
		B3	B-	B-
		Caa1	CCC+	CCC+
		Caa2	CCC	CCC
		Caa3	CCC-	CCC-
		Ca	CC	CC

		C	C	C
	Default	C	D	D

Source: CFA Program Curriculum, *Fundamentals of Credit Analysis*

- Bonds rated AAA/Aaa have the highest quality, lowest credit risk, and lowest probability of default. They have extremely low default rates.
- Bonds rated Baa3/BBB- or higher are investment grade bonds. Investment grade is further divided into high-quality, upper-medium and low-medium.
- Bonds rated lower than Baa3/BBB- are non-investment grade bonds. Non-investment grade includes speculative and default bonds. These have higher default risk.
- Outlook: This is an additional indicator on a rating. The outlook can be *positive, stable, or negative*.
- The rating agencies may also indicate the potential direction of the ratings by including comments like “on review for a downgrade” or “on credit watch for an upgrade.”
- ‘Credit migration risk’ is the risk that a bond issuer’s creditworthiness deteriorates, or migrates to a lower rating, leading investors to believe that the risk of default is higher.

Note: “Investment-grade ratings focus more on timeliness, while non-investment grade ratings give more weight to recovery”.

Credit Rating Considerations

There is sufficient evidence that rating agencies, in general, have done a good job of assigning ratings that reflect the risks involved. However, these ratings are not foolproof and have their own limitations. Analysts should conduct their own credit analysis, as sole reliance on credit rating to make investment decisions has several pitfalls:

- Credit ratings can be very dynamic: Credit rating of a bond issue may change over time. For instance, a bond may have had “AAA” rating when it was issued. But five years later, it may be A. There have been instances where an “AAA” rated bond at issuance has turned into non-investment grade or junk within three years.
- Rating agencies are not infallible: The high ratings on subprime-backed mortgage securities are the best example of this. If the rating agencies had rated the issue appropriate to their risk level, it would have made investors cautious.
- Idiosyncratic or event risks are difficult to capture in ratings: Rating agencies do a poor job of foreseeing event risks because it is not possible and these cannot be reflected in a credit rating. A recent example is of the March 2011 tsunami that hit Japan, which in turn affected the credit risk of Tokyo Electric Power Company.
- Ratings tend to lag market pricing of credit: Bond prices move on a daily basis based on the perceived credit risk in them, as the market sees it. But rating agencies do not revise their ratings that often. Investors who time their buy/sell decisions solely on the ratings issued by agencies are at the risk of underperforming the market. But a

sudden downgrade has a disastrous effect on bond prices.

4. Factors Impacting Yield Spreads

The yields on extremely liquid bonds with virtually no default risk (e.g., government bonds) will be equal to real interest rate plus a premium for expected inflation.

The yields on corporate bonds will include an additional risk premium that compensates investors for credit and liquidity risk and possibly the tax impact of holding a specific bond. This additional risk premium is called a spread.

Factors that affect the spread on corporate bonds:

- Credit cycle: Credit spreads narrow when the credit cycle improves; a spread is narrower at the top of a credit cycle. Credit spreads widen when the credit cycle deteriorates as seen during the recent credit crisis after 2008.
- Broader economic conditions: Credit spreads widen in a weak economy and narrow in a strong economy.
- Funding availability in the financial sector: Unlike stocks that primarily trade on exchanges, bonds usually trade over the counter. Brokers and dealers are market makers in the debt market. Yield spreads are narrow when brokers-dealers provide sufficient capital. However, at times of crisis as seen during the post-2008 period, many broker-dealer firms either closed down or were acquired. The capital available for making markets reduced substantially leading to wider spreads.
- General market supply and demand: If the supply is more and demand is less, credit spreads will widen. Conversely, if the demand is high and supply is low, then spreads tighten.
- Financial performance of the issuer: Earnings releases, news, and other developments associated with the issuer affect credit spreads. Good news will narrow spreads while bad news will widen spreads.

A number of these factors caused spreads to widen dramatically during the 2008-2009 global financial crisis.

Reasons for investing in HY bonds

Aside from the higher coupons typically offered by high-yield bonds to compensate for their higher risk, there are several reasons to invest in HY bonds:

- Portfolio diversification: HY bonds typically have a low correlation with IG bonds and default risk-free interest rates. Their inclusion in a fixed-income portfolio may therefore provide diversification benefits.
- Capital appreciation: Economic recovery or improved issuer-specific performance has a more sizeable positive impact on HY than IG bond prices.
- Equity-like return with lower volatility: While HY bond and equity prices tend to fluctuate similarly over the economic cycle, HY bond returns tend to be more stable

due to their higher income component versus stocks.

The Price Impact of Spread Changes

How changes in spread affect the price and return of a bond:

Two factors that affect the return on a bond are:

- Modified duration: Price sensitivity to interest rate changes.
- Magnitude of the spread change.

Return impact is calculated using these formulae:

Return impact with and without convexity adjustment

For small changes in yield:

Return impact (without convexity adjustment) $\approx -\text{ModDur} \times \Delta\text{spread}$

For large changes in yield:

Return impact (with convexity adjustment) $\approx -(\text{ModDur} \times \Delta\text{spread}) + \frac{1}{2} \times \text{Convexity} \times (\Delta\text{spread})^2$

Note: The negative sign indicates that bond prices and yields move in opposite directions.

For option-free bonds, convexity should be scaled so it has the same order of magnitude as duration squared and the spread change is expressed as a decimal. For example, if a bond has duration of 5.0 and reported convexity of 0.235, then first re-scale convexity to 23.5, and then apply the formula. For a 1% increase in spread, the result would be

$$\% \Delta PV_{\text{Full}} = (-5.0 \times 0.01) + \frac{1}{2} \times 23.5 \times (0.01)^2 = -0.048825 \text{ or } -4.8825\%.$$

Example:

(This is Question Set- Q#4 from the curriculum.)

For a bond with a modified duration of 4 and a convexity of 0.25, which of the following changes in credit spread would result in a price decrease closest to 7.5%?

- A. 1% decrease
- B. 1% increase
- C. 2% increase

Solution:

The correct answer is C, as shown by solving for ΔSpread in Equation 4:

$$\begin{aligned} \% \Delta PV_{\text{Full}} &= -(\text{AnnModDur} \times \Delta\text{spread}) + \frac{1}{2} \text{AnnConvexity} \times (\Delta\text{spread})^2 \\ &= -(4 \times 0.02) + \frac{1}{2}(25) \times (0.02)^2 = -0.075 \text{ or } -7.5\%. \end{aligned}$$

The spread change is inversely related to the price effect, with a spread increase leading to a fall in bond price. Note that since duration was 4, we had to rescale the convexity from 0.25

to 25.

Summary

LO: Describe credit risk and its components, probability of default and loss given default.

Credit risk is the risk of loss if the borrower fails to make scheduled payments of interest and/or principal.

Credit risk has two components:

- Probability of Default (POD): The probability that the borrower will default.
- Loss given Default (LGD): If the borrower does default, how severe is the loss.

Expected Loss = $POD \times LGD$

LO: Describe the uses of ratings from credit rating agencies and their limitations.

Credit rating agencies, such as Moody's, Standard & Poor's, and Fitch issue credit ratings for bonds which are opinions about a bond issue's creditworthiness. These ratings help investors compare the credit risk of different debt issues.

The risks of relying on ratings from credit rating agencies are:

- Credit ratings can be very dynamic.
- Rating agencies are not foolproof.
- Idiosyncratic or event risks are difficult to capture in the ratings.
- Ratings tend to lag market pricing of credit.

LO: Describe macroeconomic, market, and issuer-specific factors that influence the level and volatility of yield spreads.

Factors that affect the spread on corporate bonds:

- Credit cycle: Credit spreads narrow when the credit cycle improves; it is narrower at the top of a credit cycle.
- Broader economic conditions: Credit spreads widen in a weak economy and narrow in a strong economy.
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